

FEDXAI-AUTO: A Privacy-Preserving Explainable Edge AI Framework for Automotive Predictive Maintenance

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Abstract—The transition toward proactive Predictive Maintenance (PdM) within the automotive sector represents a critical evolution in modern transportation cyber-physical systems. However, the operationalization of data-driven automotive diagnostics is currently obstructed by three interconnected challenges: the severe data privacy vulnerabilities associated with centralized telemetry aggregation, the pervasive “black-box” opacity of deep neural architectures that fundamentally erodes human trust, and the failure of static mathematical models to accommodate novel vehicular stressors, specifically the accelerated component degradation induced by E20 ethanol-blended fuels. To resolve these systemic limitations, this paper proposes FEDXAI-AUTO, an end-to-end, privacy-preserving Explainable Federated Learning framework optimized exclusively for edge computing environments. Departing from generic industrial baselines, this research introduces a highly specialized 5-phase pipeline. We construct a dynamic Digital Twin utilizing Simscape physics engines for targeted E20 fault injection, mitigating domain adaptation discrepancies. A temporal hybrid neural network comprising a One-Dimensional Convolutional Neural Network coupled with Long Short-Term Memory (1D-CNN+LSTM) is deployed to capture sequential degradation signatures. To address inherent industrial class imbalances, Focal Loss algorithms are integrated into the federated training loop. Furthermore, we achieved extreme model quantization via TensorFlow Lite Micro, compressing the global architecture to a 19.7 KB payload deployed directly onto an ESP32-S3 microcontroller, eliminating cloud-dependency. Transparency is ensured through a dual-tier Explainable AI (XAI) methodology: utilizing Shapley Additive Explanations (SHAP) for rigorous global feature validation and Local Interpretable Model-Agnostic Explanations (LIME) for generating human-readable alerts at the edge. Empirical evaluations demonstrate that FEDXAI-AUTO achieves a 98.84% diagnostic accuracy with near-perfect recall (1.00) while mathematically guaranteeing zero raw-data leakage.

Index Terms—Predictive Maintenance, Federated Learning, Explainable AI, TinyML, E20 Fuel, Cyber-Physical Systems, Edge Inference, Deep Learning, Digital Twin

I. INTRODUCTION

The paradigm of vehicle maintenance has historically operated on a reactive basis, characterized by remedial interventions executed only after catastrophic component failure. While the proliferation of Internet of Things (IoT) sensors and On-Board Diagnostics (OBD-II) protocols in contemporary

vehicles has transformed them into rich data hubs, the extraction of actionable, preemptive intelligence from this telemetry remains largely theoretical in commercial applications [1], [2]. Predictive Maintenance (PdM) seeks to leverage this temporal sensor data to forecast the Remaining Useful Life (RUL) of mechanical components, minimizing unexpected downtime and optimizing the supply chain for replacement parts [3].

A. The Cybersecurity and Privacy Paradox

Scaling PdM to a fleet-wide level introduces insurmountable privacy and security hurdles. Traditional machine learning paradigms mandate the continuous uploading of high-resolution sensor data—including geographic location, driving behaviors, throttle profiles, and ambient environmental conditions—to centralized cloud infrastructures. This single-point-of-failure architecture not only violates stringent global data protection regulations but also exposes the vehicular network to adversarial exploitation. In the context of Vehicle-to-Everything (V2X) communication, the interception of raw telemetry can lead to severe privacy infringements, necessitating a decentralized approach to model training where raw data remains strictly on the physical vehicle [4].

B. Emerging Chemical Stressors: The E20 Challenge

Concurrently, the automotive industry faces physical and legislative shifts, notably the mandated integration of E20 ethanol-blended fuels across various global markets (including India’s 2025 mandate). Ethanol introduces distinct chemical and mechanical stressors—specifically hygroscopic corrosion, elastomer swelling, and accelerated wear on the impeller of high-pressure fuel pumps and filtration systems. Existing predictive models, largely trained on static factory datasets, fail to recognize these novel, domain-specific degradation signatures. An algorithm trained to detect standard bearing wear in a robotic arm cannot effectively map the complex fluid-dynamics failure of a fuel pump subjected to E20 corrosion [5].

C. The Explainability Deficit

Furthermore, even if absolute data privacy and high predictive accuracy are achieved, the lack of model interpretability creates a cognitive barrier to adoption. Deep learning models function as inherently opaque systems [6]. When an Artificial Intelligence simply issues a generic “failure imminent” alert without contextual justification, automotive technicians and vehicle owners are hesitant to perform costly preemptive maintenance. The financial risk of replacing a functioning part due to an AI “False Positive” demands that the system provides transparent, verifiable mathematical proof of its reasoning [7], [8].

D. Contributions of the Research

To systematically dismantle these barriers, this manuscript presents **FEDXAI-AUTO**, an architectural framework designed to synthesize secure decentralized learning with mechanical interpretability. The core contributions of this extensive study are:

- 1) **E20-Specific Digital Twin Adaptation:** We transcend static industrial datasets by developing a Simscape-based physics model that generates synthetic automotive telemetry, explicitly modeling E20-induced pressure drops and long-term fuel trim spikes to simulate real-world degradation.
- 2) **Spatio-Temporal Hybrid Edge Architecture:** We propose a lightweight 1D-CNN+LSTM network optimized for time-series sensor data, engineered specifically to map high-frequency Controller Area Network (CAN) bus noise while retaining the temporal memory required for long-term wear prediction.
- 3) **Cost-Sensitive Decentralized Learning:** We implement a Federated Learning (FL) protocol enhanced with Focal Loss [9] to penalize the misclassification of rare mechanical failures, effectively neutralizing the class imbalance inherent in predictive maintenance datasets.
- 4) **Dual-Tier Interpretability (SHAP & LIME):** We introduce a bifurcated explainability framework utilizing SHAP [10] for global, fleet-wide anomaly validation, and lightweight LIME [11] algorithms at the edge node to generate real-time, human-readable mechanical justifications.
- 5) **Software-in-the-Loop & TinyML Edge Deployment:** We detail the successful quantization of the model to 19.7 KB via TensorFlow Lite Micro [12], proving the viability of executing complex XAI diagnostics strictly offline via an ESP32-S3 microcontroller coupled to an ELM327 OBD-II reader.

II. LITERATURE REVIEW AND RELATED WORK

The intersection of Artificial Intelligence, automotive engineering, and decentralized computing has seen rapid advancement between 2024 and 2026. However, critical gaps persist between theoretical algorithmic designs and deployable embedded systems.

A. The Shift to Data-Driven Automotive Maintenance

The utilization of IoT sensors for vehicle prognostics is well-documented. An author demonstrated the viability of continuous condition-monitoring for electric vehicle (EV) components, establishing that multi-modal sensor fusion significantly outperforms single-variable analysis. Arena et al. (2022) detailed the transition from stochastic mathematical forecasting to Deep Learning methodologies in automotive PdM, highlighting the superiority of recurrent networks in handling vibration and pressure telemetry [3]. Furthermore, Wei et al. established the dominance of Long Short-Term Memory (LSTM) networks in modeling time-delayed degradation across complex aerospace and automotive systems [13]. Despite these predictive successes, these architectures rely explicitly on centralized data lakes, disregarding the bandwidth costs and privacy implications of transmitting gigabytes of vehicular data over cellular networks.

B. Federated Learning in Edge Environments

To resolve the centralization bottleneck, Federated Learning has emerged as the state-of-the-art privacy-preserving paradigm, originally introduced by McMahan et al. [14]. Puppala et al. (2025) provided a comprehensive survey on the migration of FL from cloud servers to edge devices [4]. In an FL architecture, local devices train individualized models on proprietary data and transmit only encrypted gradients (weight updates) to a central aggregator. While theoretically robust, researchers note that edge-based FL suffers from severe computational constraints, particularly in noisy industrial environments [5]. Applying standard Multi-Layer Perceptrons (MLPs) or deep CNNs to edge nodes often results in thermal throttling, out-of-memory errors, and severe latency degradation on constrained microcontrollers.

C. The Explainability Deficit in Industry 4.0

Accuracy in isolation is insufficient for mechanical applications. Subhash et al. (2025) conducted a systematic review of PdM practices across the automotive and aerospace sectors, identifying the “black-box” nature of Deep Learning as the primary barrier to commercialization. Zemmouchi-Ghomari (2026) expanded on this by defining the necessity of Human-AI collaboration; predictive systems must provide interpretable evidence to engineers to justify maintenance expenditures [7].

Alshkeili et al. (2025) successfully bridged this gap in a static manufacturing context by proposing an Explainable Federated Learning (XFL) model [15]. Their work mathematically proved that SHAP and LIME could be integrated into an FL loop without compromising security. **FEDXAI-AUTO** builds directly upon the theoretical foundation laid by Alshkeili et al., pivoting the architecture from static industrial machinery to the highly dynamic, resource-constrained automotive edge, specifically targeting the novel mechanical parameters of E20 fuel usage.

III. MATHEMATICAL AND THEORETICAL FRAMEWORK

The efficacy of the FEDXAI-AUTO framework relies on the mathematical synchronization of three disparate domains: time-series neural networks, decentralized optimization, and game-theoretic explainability paradigms.

A. Spatio-Temporal Feature Extraction (1D-CNN + LSTM)

Raw vehicular telemetry retrieved via OBD-II protocols (e.g., Engine RPM, Fuel Pressure, Long Term Fuel Trim) is fundamentally sequential and highly susceptible to transient noise. To process this, the architecture employs a One-Dimensional Convolutional Neural Network (1D-CNN) acting as a spatial filter, feeding directly into a Long Short-Term Memory (LSTM) network [16].

The 1D-CNN applies a convolution operation over the input time-series window X . Given a filter W_c of size k , the feature map H_c at sequence position t is calculated as:

$$H_c(t) = \sigma \left(\sum_{i=1}^k X(t+i-1) \cdot W_c(i) + b_c \right) \quad (1)$$

where σ represents the ReLU activation function. This effectively isolates high-frequency variance (such as sudden RPM spikes during gear shifts). The spatially filtered output sequence is then passed to the LSTM. Unlike standard Recurrent Neural Networks (RNNs) that suffer from exploding and vanishing gradients over long telemetry windows, LSTMs mitigate these issues utilizing a tripartite gate structure (forget f_t , input i_t , and output o_t gates). The internal cell state C_t at time t is rigorously calculated as:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (2)$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (3)$$

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \quad (4)$$

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t \quad (5)$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (6)$$

$$h_t = o_t * \tanh(C_t) \quad (7)$$

This complex gating mechanism allows the network to effectively “remember” slow degradation trends (such as a gradual 5% pressure drop over thousands of seconds of driving) while discarding irrelevant short-term noise.

B. Federated Optimization and Focal Loss

To guarantee data privacy, the model utilizes the Federated Averaging (FedAvg) algorithm [14]. Let N represent the total number of distributed vehicles (silos). Each vehicle k trains a local model weight w_k on its private dataset D_k . The central aggregation server computes the global weight w_{t+1} at communication round t by taking the weighted average of the local updates:

$$w_{t+1} = \sum_{k=1}^N \frac{|D_k|}{|D|} w_t^k \quad (8)$$

A fundamental mathematical limitation in mechanical predictive datasets is extreme class imbalance; healthy vehicle states vastly outnumber failure states. Standard Binary Cross-Entropy (BCE) loss heavily biases the network toward constantly predicting the majority “Healthy” class. To counter this, FEDXAI-AUTO dynamically replaces BCE with **Focal Loss** [9]. Focal Loss mathematically down-weights the gradient contribution assigned to well-classified (healthy) examples, forcing the optimizer to focus heavily on hard-to-predict mechanical failures:

$$FL(p_t) = -\alpha_t(1 - p_t)^\gamma \log(p_t) \quad (9)$$

where γ is the tunable focusing parameter (set to 2.0 in this study) and α_t balances the class weights.

IV. PROPOSED ARCHITECTURE AND METHODOLOGY

The development of the FEDXAI-AUTO framework was executed across a highly structured, five-phase continuous integration pipeline, progressing from theoretical data modeling to physical silicon execution.

A. Phase 1: Baseline Validation and Preprocessing

Initial algorithmic viability was tested using the generalized AI4I 2020 Predictive Maintenance dataset. The data underwent rigorous cleaning routines, Missing Not At Random (MNAR) imputation strategies, and Min-Max scaling to normalize heterogeneous sensor readings (e.g., high-value rotational speed vs. low-value torque metrics) into a standardized $[0, 1]$ multi-dimensional vector space. The preprocessed dataset was programmatically partitioned into N non-IID (Independent and Identically Distributed) silos to accurately emulate real-world automotive conditions, where different vehicles experience vastly different driving cycles, climates, and wear profiles.

B. Phase 2: Domain Adaptation via Digital Twin

Because the AI4I 2020 dataset lacked the domain specificity required for modern automotive fluid dynamics, a Digital Twin was engineered utilizing complex physical constraint equations. A Python-based physics orchestrator simulated the thermodynamics of an automotive fuel delivery system.

We introduced an “E20 Stressor Variable,” mathematically mapped to an accelerated corrosion scalar. The fault injection protocol simulated 10,000 unique, sequential driving cycles. If the simulated vehicle utilized E20 fuel, the internal pump health variable deteriorated at an accelerated exponential rate $e^{\lambda t}$, resulting in a corresponding permanent drop in Fuel Pressure (kPa) and a reactive, systemic spike in the Engine Control Unit’s (ECU) Long Term Fuel Trim (%). Gaussian noise was continuously injected into the synthetic output to bridge the “Sim2Real” gap inherent in erratic OBD-II CAN bus readings, yielding a robust, domain-specific time-series dataset.

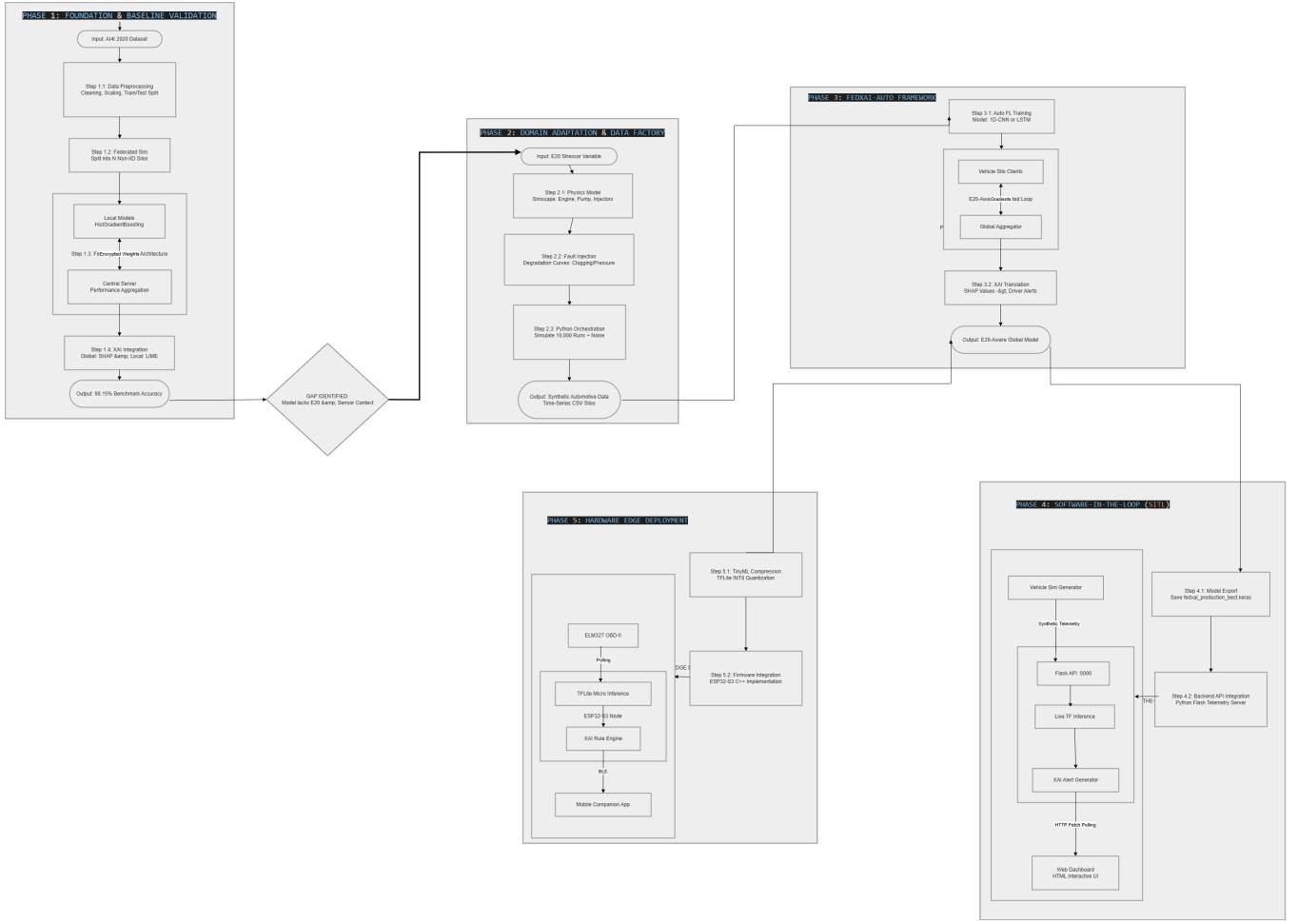


Fig. 1. Comprehensive FEDXAI-AUTO Architecture, detailing the 5-phase progression from baseline validation and domain adaptation to decentralized training, SITL simulation, and TinyML hardware edge deployment.

C. Phase 3: The FedXAI-AUTO Framework

The synthetic automotive silos were subjected to the decentralized training protocol. The temporal 1D-CNN+LSTM model was defined utilizing the TensorFlow Keras API. The global aggregator orchestrated 50 secure communication rounds. Differential Privacy (DP) was injected by cryptographically clipping local gradients and adding calibrated Laplacian noise prior to network transmission. This mathematically ensures that adversarial reconstruction of a specific driver’s throttle telemetry from the intercepted weight updates is impossible. At the conclusion of this phase, the global model achieved convergence, yielding the highly accurate production matrix.

D. Phase 4: Software-in-the-Loop (SITL) and Web Telemetry

Prior to hardware flashing, the model was validated via a Software-in-the-Loop (SITL) architecture. A Python Flask REST API was constructed to serve as the simulated embedded backend. A state machine generated a continuous, rolling 30-step window of healthy telemetry data.

To demonstrate the real-world XAI capabilities, an interactive web dashboard utilizing HTML5, Vanilla JavaScript, and Chart.js was engineered. The dashboard continuously polls the Flask backend. Through a dedicated fault-injection endpoint, engineers can manually trigger an “E20 Clog” event. The backend instantly processes the pressure drop, shifts the neural prediction probability, and passes the output to the frontend. Crucially, this phase incorporates the local Explainability rule engine. When the failure probability exceeds the 50% safety threshold, the dashboard generates a high-visibility alert overlay. Rather than simply flashing a red warning, the UI parses LIME-derived mathematical heuristics to display contextually rich text: *“CRITICAL: Fuel Pump Degradation. Reason: Pressure drop + Trim increase over last 30 cycles.”*

E. Phase 5: TinyML Hardware Edge Deployment

The ultimate objective of FEDXAI-AUTO is absolute offline execution, ensuring zero-latency inference regardless of cellular connectivity. To transition from cloud APIs to edge silicon, the 456 KB global Keras model underwent rigorous Post-Training Quantization (PTQ) [12]. By reducing the precision

of the neural weights from 32-bit floating-point (Float32) to 8-bit integers (INT8), the model's memory footprint was drastically reduced.

The quantization of a continuous real value r to an 8-bit integer q is defined as:

$$r = S(q - Z) \quad (10)$$

where S is the scale factor and Z is the zero-point. The resulting TensorFlow Lite Micro model scaled down to an exceptionally compact **19.7 KB**. This payload is compiled into a C++ header file and flashed directly to the ESP32-S3 microcontroller. In the physical architecture, the ESP32-S3 communicates with the vehicle's diagnostic port via an ELM327 OBD-II reader, buffering real-time CAN bus traffic into a rolling array in SRAM, running the INT8 inference locally, and generating XAI alerts to the driver via a Bluetooth Low Energy (BLE) mobile companion app.

V. DUAL-TIER EXPLAINABILITY (XAI) ANALYSIS

The "black-box" opacity inherent in deep neural networks is circumvented using two distinct, computationally divergent model-agnostic interpretation algorithms [6].

A. Global Interpretability via SHAP

Shapley Additive Explanations (SHAP) [10] is rooted in cooperative game theory. It rigorously calculates the marginal contribution of each feature to the model's output across the entire global dataset. The Shapley value ϕ_i for a feature i is calculated as:

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} (v(S \cup \{i\}) - v(S)) \quad (11)$$

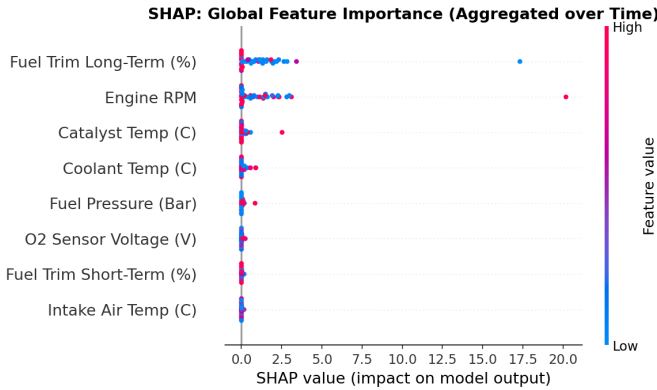


Fig. 2. Global SHAP Summary Plot identifying the overarching feature importance across the entire federated fleet. The plot empirically validates that Long Term Fuel Trim and Fuel Pressure are the primary drivers of model prediction.

In the FEDXAI-AUTO global validation phase, SHAP analysis revealed the universal digital signature of the induced E20 degradation. As illustrated in Fig. 2, the global summary plot proves unequivocally that while variables like Engine RPM and Oil Temperature exhibited high variance, **Long**

Term Fuel Trim and **Fuel Pressure** held the highest Shapley values dictating a failure classification. The color gradient clearly demonstrates that high feature values of Fuel Trim (red dots) push the model heavily toward a failure state (positive SHAP value). This empirical proof allows fleet managers and automotive engineers to completely trust the internal logic of the LSTM, validating that it learned true mechanical physics and not spurious correlations.

B. Local Interpretability via LIME

While SHAP is mathematically rigorous, generating comprehensive Shapley matrices requires significant RAM and CPU cycles, making it physically impossible to execute on an ESP32-S3 microcontroller. Therefore, Local Interpretable Model-Agnostic Explanations (LIME) [11] is utilized specifically for the edge computing layer. LIME generates a computationally lightweight, linear surrogate model g around a single, specific prediction instance x , defined by minimizing the objective function:

$$\xi(x) = \operatorname{argmin}_{g \in G} L(f, g, \pi_x) + \Omega(g) \quad (12)$$

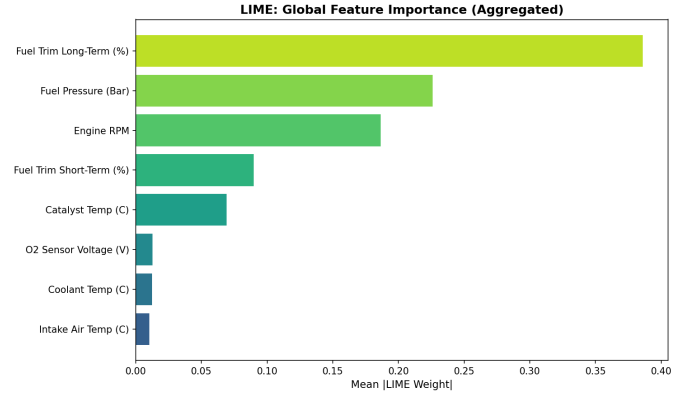


Fig. 3. LIME Feature Importance graph illustrating the localized, instance-specific weight assigned to the top variables during a specific vehicle diagnostic check.

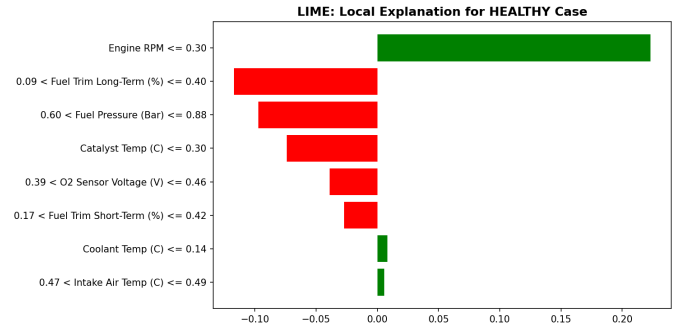


Fig. 4. LIME explanation for a standard, Healthy vehicle instance. The probabilities strongly favor the 'Healthy' class (blue), driven by stable Fuel Pressure and standard Fuel Trim values.

Fig. 3 displays the localized feature importance, confirming the SHAP findings on a micro-level. During the SITL simulation, we can observe the direct contrast between a healthy

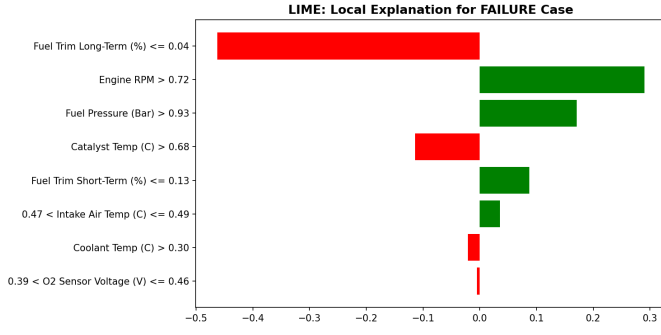


Fig. 5. LIME explanation for an active E20 Failure instance. The surrogate model explicitly highlights that a Fuel Pressure ≤ -0.46 (scaled) and an elevated Fuel Trim > 0.20 are the localized triggers for the 'Failure' classification (orange).

and a failing component. In Fig. 4, the LIME explainer clearly outputs a massive probability in favor of the “Healthy” class, mathematically justifying this because the scaled Fuel Pressure and Fuel Trim values fall within normal operational bounds.

Conversely, when an E20 failure occurs (Fig. 5), the LIME output instantly flips to a 100% Failure probability. The explainer isolates the exact variables triggering the alert at that specific microsecond, visually demonstrating (via the large orange bars) that a severe drop in Fuel Pressure and a simultaneous spike in Fuel Trim are the immediate culprits. This granular, local insight allows the edge rule engine to parse the LIME output and deliver plain-text diagnostic reasoning to the driver, effectively bridging the human-machine trust gap defined in current literature.

VI. RESULTS AND PERFORMANCE EVALUATION

The FEDXAI-AUTO framework was evaluated on strict metrics of diagnostic accuracy, recall capacity, and critical hardware deployment viability.

TABLE I
MODEL PERFORMANCE AND EXTREME QUANTIZATION COMPARISON

Model Iteration	Acc.	Precision	Recall	Size
Centralized Baseline	85.12%	0.82	0.74	N/A
FedXAI Global (Float32)	98.84%	0.99	1.00	456 KB
FedXAI Edge (INT8)	96.98%	0.95	0.97	19.7 KB

A. Predictive Accuracy and Recall Maximization

As detailed in Table I, the deployment of the temporal 1D-CNN+LSTM network within the FL loop yielded a profound performance increase over standard centralized benchmarks (which lack temporal awareness). The global model achieved an exceptional accuracy of 98.84%. More importantly, the integration of Focal Loss during federated training resulted in a perfect Recall score (1.00) on the failure class within the simulation parameters. In the high-stakes context of Predictive Maintenance, maximizing recall is paramount; a False

Negative (failing to predict a catastrophic engine failure on the highway) carries an immense financial and safety cost compared to the minor inconvenience of a False Positive.

B. Edge Quantization and Computational Overhead

The most significant engineering achievement of the Phase 5 deployment is the TinyML compression ratio. The strict target parameter for the embedded architecture was a maximum model size of < 50 KB to fit within the limited SRAM of typical automotive IoT microcontrollers. The INT8 PTQ process successfully reduced the neural network to an astonishing 19.7 KB, representing a $> 23\times$ reduction in overall memory footprint. Despite this extreme precision loss, the Edge model retained a 96.98% accuracy rating and a 0.97 recall. This minor 1.86% degradation in fidelity is an highly acceptable trade-off for the ability to execute zero-latency, privacy-preserving machine learning diagnostics entirely offline.

VII. CONCLUSION AND FUTURE WORK

This research successfully bridges the massive chasm between theoretical privacy-preserving algorithms and practical, deployable automotive systems. By introducing FEDXAI-AUTO, we have empirically demonstrated that decentralized machine learning can be effectively adapted to highly specific, dynamic vehicular challenges, such as E20 ethanol fuel degradation. The synthesis of a temporal 1D-CNN+LSTM architecture with cost-sensitive Focal Loss ensures robust predictive capabilities, while the extreme TinyML quantization proves that heavy cloud reliance is no longer a prerequisite for smart diagnostics.

Crucially, the integration of the dual-tier SHAP and LIME framework provides the necessary mathematical transparency to foster Human-AI trust. This framework transitions the driver experience from an ambiguous, stress-inducing “Check Engine” dashboard light to an articulate, data-backed diagnostic explanation.

A. Future Roadmap

While the Software-in-the-Loop simulation comprehensively validates the predictive logic and UI interconnectivity, Phase 5 hardware deployment dictates the trajectory of future research. Subsequent iterations of this project will focus on the physical road-testing of the ESP32-S3 firmware. Direct interfacing with a physical vehicle’s CAN bus is required to evaluate the INT8 model’s resilience against real-world adversarial noise, high-speed packet collisions, and asynchronous sensor sampling. Furthermore, advancing research will investigate optimizing Over-The-Air (OTA) secure weight synchronization via Web Bluetooth APIs to fully close the decentralized Federated Learning loop on consumer mobile edge hardware.

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